

Data Quality & Preparation Notes

Insurance Claims & Risk Analysis

Business Question:

What factors are associated with higher-cost claims and longer claim resolution times?

1. Source Data

Primary Dataset

Minitab Insurance Fraud Dataset

<https://support.minitab.com/en-us/datasets/getting-started-guide-data-sets/insurance-fraud-data/>

Geographic Enrichment

U.S. ZIP Code Database

<https://simplemaps.com/data/us-zips>

The primary dataset contains insurance claim, driver, vehicle, accident, and claim-processing information. A ZIP code reference table was added to support geographic analysis by state and region.

2. Initial Data Validation

The source CSV was reviewed before database import to confirm the expected structure, column availability, and general data formatting.

Post-import validation included:

Validation Check	Result	Status
Imported row count	12,002	Passed
Duplicate claim numbers	0	Passed
Negative claim amounts	0	Passed
Future claim dates	0	Passed
Claim date range	2023-01-01 to 2024-09-09	Passed

Unrealistic driver ages	4 records	Review required
ZIP code format	Five-digit format	Passed
Column definitions/data types	Reviewed	Passed

Sample records were also reviewed after import to verify column alignment and confirm that values were loaded into the expected fields.

3. Missing Values

The source dataset uses an asterisk (*) as a missing-value indicator in several fields.

A total of 135 * values were identified across six columns, including fields such as witness presence, injury claim, claim date, claim day of week, age of vehicle, and marital status.

During data ingestion, the * values were mapped to SQL NULL values. This preserved missing observations while preventing the placeholder character from interfering with numeric and date validation.

Known missing values were cross-checked against the original CSV to confirm that the conversion was accurate.

Missing values were retained rather than automatically removed so that records were not excluded unnecessarily from the analysis.

4. Data Cleaning & Preparation

A cleaned claims_clean table was created from the imported source data.

Cleaning and standardization included:

- Standardized column names and text fields.
- Converted source missing-value markers to NULL.
- Converted claim_date from character data to the DATE data type.
- Standardized ZIP codes as text to preserve leading zeros and support geographic joins.
- Converted currency fields to appropriate two-decimal precision.

- Converted unrealistic driver ages to NULL.
- Transformed binary marital-status values into readable categories.
- Added calculated fields for claim year and claim month.
- Added a claim severity category for analysis and visualization.

The cleaned dataset was revalidated to confirm row counts, data types, and key data-quality conditions.

5. Data Quality Issues Identified

Unrealistic Driver Ages

Four records contained driver ages above 120.

No supporting demographic information was available to determine whether these values represented data-entry errors or legitimate records. Rather than inventing replacement values, the records were converted to NULL and excluded from any analyses involving driver age.

Claim Dates

Claim dates ranged from January 1, 2023 through September 9, 2024. No future-dated claims were identified.

Claim Amounts

No negative claim amounts were identified. Claim values ranged from approximately \$2.6K to \$101.8K, indicating substantial variation in claim severity across the dataset.

6. Geographic Data Enrichment

The cleaned claims data was enriched using a U.S. ZIP code reference table containing geographic information.

A LEFT JOIN was used so that all source claims would remain in the analysis dataset, including claims whose ZIP codes could not be matched to the reference table.

Before joining, ZIP codes in both datasets were standardized as text. This resolved a data-type mismatch and ensured that leading zeros were preserved.

The geographic enrichment process included:

- Validating ZIP code formatting.
- Checking for duplicate ZIP values in the lookup table.
- Assigning states to geographic regions.
- Reviewing regional assignments.
- Retaining non-standard geographic records as Other rather than deleting them.
- Measuring the ZIP code match rate.
- Creating the claims_analysis_view for downstream analysis.

Geographic Match Rate

Approximately 76.0% of claims successfully matched to the ZIP lookup table.

The relatively low match rate was investigated rather than assumed to be a SQL join error. The free ZIP reference dataset contained approximately 33,782 ZIP records and did not provide complete U.S. ZIP coverage.

The join logic was validated, and unmatched claims were retained. Therefore, the unmatched records were attributed primarily to limitations in the reference dataset rather than an error in the SQL join.

Approximately 24% of claims therefore have an Unknown geographic classification and were treated separately in geographic analysis.

7. Analysis Considerations & Assumptions

Safety Rating

The original dataset did not provide documentation defining whether higher safety-rating values represented safer or less-safe drivers.

For analysis purposes, the values were therefore ranked from lowest to highest and divided into three approximately equal-sized groups using NTILE(3).

This approach avoided imposing arbitrary thresholds without documentation from the source dataset.

The resulting groups showed very similar average claim amounts, suggesting that safety rating was not a strong individual factor associated with claim severity in this dataset.

Regional Analysis

Regional comparisons were interpreted using claim-level results rather than population-level rates. Because the dataset does not provide exposure or population information, differences in claim counts should not be interpreted as differences in population-level claim frequency.

Sample Size

Some subgroup comparisons contain substantially fewer claims than others. Results for smaller groups should therefore be interpreted cautiously and may require additional data before being used to support operational decisions.

8. Impact on Business Question

The data-quality and preparation process provided a documented basis for investigating the project's two business questions:

What factors are associated with higher-cost claims?

Individual vehicle, driver, accident, and geographic characteristics generally showed limited differences in average claim severity. Claim values varied substantially overall, suggesting that higher-cost claims may be influenced by combinations of characteristics rather than one dominant factor.

The highest-cost claims were reviewed separately to identify potential patterns. Large vehicles accounted for 6 of the top 10 highest value claims, despite representing approximately one-third of the overall dataset. However, high claim severity occurred across a broad range of vehicle values, accident locations, and regions. No single characteristic clearly explained the most expensive claims.

What factors are associated with longer claim resolution times?

Most analyzed factors—including accident site, fraud status, form defects, and region—showed relatively consistent average days open.

Claim channel showed the most notable variation: online claims averaged approximately 4 days open, compared with approximately 9–10 days for broker and phone claims. However, online claims represented a smaller share of the dataset, so this relationship should be investigated further before attributing the difference directly to the claim channel itself.

9. Key Limitations

The analysis should be interpreted within the limitations of the available data:

- The dataset does not contain all operational factors that may influence claim cost or resolution time.
- Geographic enrichment was limited to approximately 76% of claims because of incomplete ZIP reference data.
- The source dataset does not document the meaning of the safety-rating scale.
- Some subgroup analyses have smaller sample sizes.
- The analysis identifies associations, not causal relationships.
- Additional policyholder, coverage, operational, and exposure data would be needed for more comprehensive analysis.

These limitations were retained as part of the analysis rather than removing or modifying records without sufficient evidence.

10. Final Data Preparation

The final workflow produced:

Raw CSV → Validated/cleaned claims table → Geographic enrichment → Analysis view → SQL analysis → Dashboard

This process produced a cleaned and documented dataset ready for analyzing claim severity and resolution time, with key assumptions, validation checks, and data limitations recorded along the way.